

Coding & Solution

Date	15 March 2026
Team ID	XXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of the implemented solution. The project demonstrates clean coding practices, proper implementation of machine learning techniques, and a user-friendly web interface for predicting an individual's overall well-being based on health and lifestyle information.

Solution Summary

Field	Details
Repository Link / URL	(Paste your GitHub repository link here)
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy, Matplotlib
Key Features Implemented	• User-friendly web interface for entering health and lifestyle details. • Data preprocessing and cleaning. • Feature engineering and selection. • Machine Learning model for Well-Being Score Prediction. • Interactive graphs and visualizations. • Personalized health recommendations based on prediction. • Model saving and loading using Joblib.
Pending / Incomplete Features	• User authentication and profile management. • Prediction history tracking. • Email notification support. • Cloud deployment. • Advanced model optimization and hyperparameter tuning.
Setup / Run Instructions	1. Clone or download the project. 2. Install required packages using <code>pip install -r requirements.txt</code>. 3. Place the dataset in the project directory. 4. Run the application using <code>python app.py</code>. 5. Open <code>http://127.0.0.1:5000/</code> in your browser. 6. Enter health and lifestyle details to view the predicted well-being score and recommendations.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The application is developed using Python, Flask, Scikit-learn, Pandas, NumPy, HTML5, CSS3, and JavaScript to predict an individual's overall well-being based on health, lifestyle, and personal information. The solution performs data preprocessing, feature selection, model training, prediction, and visualization to generate accurate well-being scores. The application provides a responsive web interface with personalized recommendations, follows a modular software architecture, and is designed for easy maintenance, scalability, and future enhancements.